

LAMAR, CO, USA

WMO: 724636

Lat: 38.070N

Lon: 102.688W

Elev: 3704

StdP: 12.83

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 03013

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.6	(c) 6.5	(d) -6.1	(e) 4.5	(f) 8.8	(g) -1.1	(h) 5.9	(i) 17.7	(j) 28.6	(k) 42.1	(l) 25.0	(m) 43.3	(n) 6.6	(o) 210	(p) 0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 30.3	(c) 100.6	(d) 65.4	(e) 98.5	(f) 65.7	(g) 95.5	(h) 65.3	(i) 70.9	(j) 88.3	(k) 69.6	(l) 87.0	(m) 68.6	(n) 86.1	(o) 12.4	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	111.0	74.7	64.9	105.8	74.0	63.8	101.8	73.5	37.0	87.3	36.0	87.3	35.0	86.0	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.9	25.2	21.4	-9.1	105.7	7.5	2.6	-14.5	107.6	-18.9	109.1	-23.1	110.6	-28.5	112.5		
			-9.6	73.6	7.3	2.0	-14.9	75.0	-19.2	76.2	-23.4	77.3	-28.7	78.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	30.5	34.1	43.8	52.6	62.6	73.7	79.0	76.1	68.3	53.8	41.1	30.7		
	DBStd	19.25	10.30	10.95	9.81	8.94	8.31	6.99	4.97	5.10	7.80	9.07	9.14	10.07		
	HDD50	2329	606	449	237	75	9	0	0	0	2	63	288	600		
	HDD65	5321	1071	864	658	379	143	11	1	1	54	357	718	1064		
	CDD50	3774	0	5	45	152	399	711	900	810	550	181	20	1		
	CDD65	1292	0	0	1	6	68	272	436	346	153	10	0	0		
	CDH74	18637	0	9	114	415	1455	3886	5763	4180	2296	487	32	0		
	CDH80	9928	0	1	20	120	646	2150	3410	2271	1140	168	2	0		
(14) Wind	WSAvg	9.5	7.9	9.0	10.1	11.8	10.9	10.8	9.8	9.0	9.2	8.8	8.2	8.0		
(15) Precipitation	PrecAvg	15.4	0.4	0.4	0.9	1.3	2.4	2.4	2.3	2.2	1.3	0.9	0.6	0.4		
	PrecMax	22.9	1.4	2.1	4.7	5.0	7.0	6.6	7.9	7.5	3.9	4.8	2.0	1.7		
	PrecMin	9.6	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.1	0.0	0.0	0.0	0.0		
	PrecStd	3.4	0.4	0.5	0.8	1.2	1.7	1.6	1.5	1.7	1.0	0.9	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	74.1	82.4	88.3	96.6	105.3	103.9	101.9	99.2	89.9	78.3	67.5		
		MCWB	45.4	48.1	52.2	55.3	60.6	62.8	66.6	65.3	64.4	59.0	51.8	46.7		
	2%	DB	61.3	67.1	77.2	83.5	91.4	99.9	101.1	98.6	94.9	84.7	72.4	60.7		
		MCWB	42.8	45.5	50.0	53.6	59.6	63.6	66.3	66.3	63.2	56.8	49.4	42.8		
	5%	DB	54.8	61.2	71.7	79.2	87.5	96.2	99.0	95.3	91.1	80.0	66.2	54.5		
		MCWB	40.1	42.9	48.8	52.7	59.4	63.6	66.7	66.2	62.8	55.9	47.3	40.3		
	10%	DB	48.1	54.2	65.2	73.4	82.5	91.8	96.4	91.8	87.6	73.4	60.1	48.2		
		MCWB	37.2	39.8	46.3	51.5	58.2	63.5	67.0	66.6	62.4	53.7	44.4	37.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	49.3	55.0	59.2	66.2	71.1	72.5	72.4	69.0	62.8	54.1	47.6		
		MCDB	64.2	70.7	75.2	75.9	84.9	87.5	90.4	88.3	85.7	79.7	71.0	65.6		
	2%	WB	43.5	46.2	51.9	56.7	63.9	68.9	71.0	70.9	67.2	59.8	50.8	43.8		
		MCDB	60.5	65.3	71.7	75.6	81.5	86.2	89.1	87.6	84.6	76.4	68.3	59.2		
	5%	WB	40.6	43.5	49.6	54.6	62.0	67.4	69.9	69.5	65.7	57.5	48.3	40.7		
		MCDB	54.1	60.3	68.7	73.1	80.0	85.2	88.1	85.7	84.0	74.6	65.2	53.8		
	10%	WB	37.5	40.4	47.1	52.5	60.2	66.0	68.9	68.4	64.1	55.2	45.3	37.7		
		MCDB	47.1	53.3	63.7	70.5	77.6	84.0	87.2	84.9	81.4	71.6	59.3	48.0		
(35) Mean Daily Temperature Range	MDBR		30.1	30.4	33.1	31.5	31.6	31.4	30.3	29.0	31.4	33.1	32.6	30.1		
	5% DB	MCDBR	40.7	42.9	43.9	42.4	38.9	37.3	35.4	34.2	37.6	42.6	43.5	39.7		
		MCWBR	24.4	23.8	21.2	18.6	15.1	11.6	9.8	9.8	12.9	19.1	22.8	24.1		
	5% WB	MCDBR	38.3	40.7	40.4	36.2	33.8	31.7	29.5	29.1	32.2	36.3	40.8	38.1		
		MCWBR	23.4	22.9	20.3	16.9	14.7	11.9	9.6	10.3	11.9	17.5	22.2	23.7		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.258	0.285	0.315	0.333	0.350	0.373	0.360	0.325	0.285	0.261	0.250		
	taud		2.583	2.574	2.496	2.423	2.435	2.416	2.404	2.405	2.489	2.580	2.569	2.576		
	Ebn at Noon		305	314	312	306	300	293	285	288	294	299	297	297		
	Edh at Noon		23	26	31	36	36	37	37	36	31	26	23	21		
(44) All-Sky Solar Radiation	RadAvg		867	1128	1561	1922	2216	2415	2380	2104	1751	1307	962	762		
	RadStd		64	103	123	102	120	101	80	85	87	107	69	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	N/A

Nomenclature: See separate page